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EE695

SIMULATION OF LOW EARTH ORBIT CONSTELLATIONS IN RESPECT TO RURAL BROADBAND COVERAGE IN IRELAND

A Project Report submitted in partial fulfilment of the requirements of the Maynooth University for the degree of Masters of Engineering

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Abstract

Declaration

1. Introduction

Reliable broadband has become fundamental for modern life, enabling access to information, communication, education, and economic opportunities. However, many rural areas in Ireland still lack access to high-speed broadband connectivity, creating a significant digital divide between urban and rural communities. This disparity limits the ability of rural residents to fully participate in the digital economy and access essential online services.

In Ireland, this divide can be particularly pronounced, despite initiatives such as the NationaL BroadBand Plan (NBBP) which aims to provide high-speed broadband to all premises by 2028. Large Geographical areas with low population density, such as the west of Ireland, present significant challenges for traditional terrestrial broadband deployment due to high infrastructure costs and logistical difficulties. As a result, many rural communities remain under-served, with limited options for reliable internet access. As a result, technologies capable of delivering high-speed, low-latency, broadband without extensive ground infrastructure should be explored.

Low Earth Orbit (LEO) satellite systems have emerged as a potential solution to these challenges. Operating at altitudes significantly lower than geostationary satellites, LEO constellations can offer reduced latency, improved link budgets and more flexible coverage. This project aims to evaluate the potential of LEO satellite systems to provide broadband connectivity to rural areas in Ireland. By modelling the coverage and performance of existing and planned LEO constellations, this project seeks to assess their ability to bridge the digital divide and provide reliable internet access to under-served communities.

This project will address these challenges by developing a simulation to evaluate the coverage and capacity potential of LEO satellite systems over Ireland, using publicly available orbital data and established propagation models, the work will quantify satellite coverage times, and theoretical data rates across a grid of locations in Ireland. The results will provide insights into the feasibility of LEO satellite systems as a solution for rural broadband connectivity.

1.1 Background and Motivation

Terrestrial network deployment, particularly fibre based networks can be costly, and typically more cost effective in more urban areas where return on investment is more apparent. Rural regions by comparison have dispersed populations and potentially challenging terrain which can significantly increase the deployment costs, limiting infrastructure expansion.

In Ireland this difference is evident from Figure 1.1, where the areas in Amber are targets for state intervention under the National Broadband Plan (NBP) and the areas in blue are areas where commercial operators ae already delivering high speed broadband services [1]. It is evident through this map that areas such as Dublin receive significantly improved service.

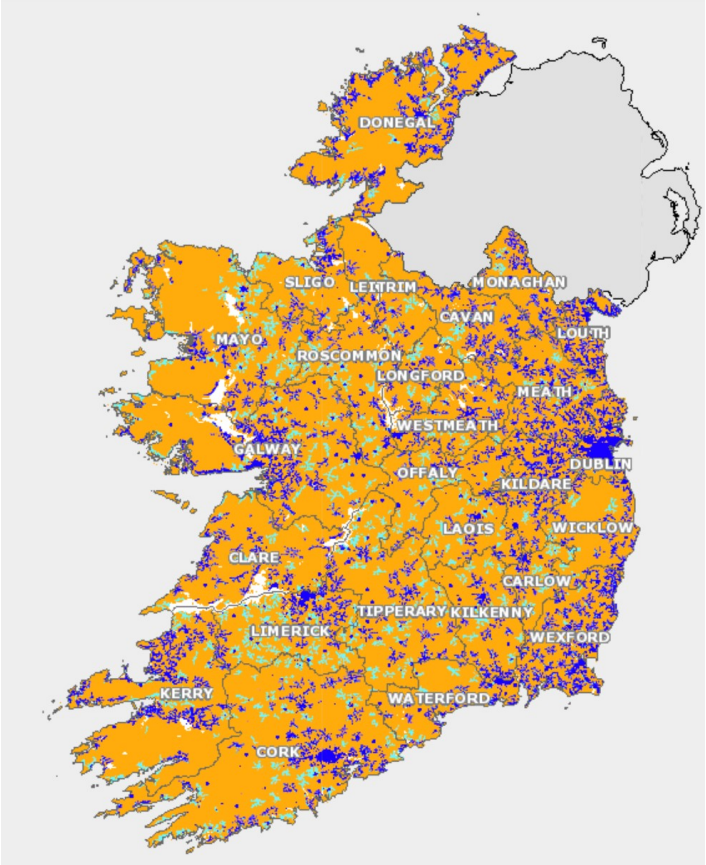


Figure 1.1: High speed broadband in parts of Ireland [1]

1.2 What is LEO

LEO satellites orbit closely to the Earth's surface, this can be anywhere below $\approx 1500\,\text{km}$, however some even orbit as low as 160km. These satellites do not have to follow a particular orbit and are not locked in one spot above the Earth. LEO satellites have a multitude of uses such as satellite imaging of the Earth due to the distance of orbit they are able to take images at a higher resolution. The Internation space station (ISS) is also in low earth orbit, this means it is cheaper and easier for the astronauts to travel to and from it. This report is largely concerned with LEOs ability to send data quickly back to Earth due to it's close proximity. Often times these satellites work as part of a constellation in order to achieve constant coverage over an area [2].

This is slightly different from other types of satellites such as Geostationary orbit (GEO), medium Earth orbit satellites (MEO). GEO satellites match Earth's rotation, taking a full 23 hours and 56 minutes to orbit. These satellites are locked into place at a fixed orbit, traveling around 3 kilometers per second at an altitude of 35786 Km. GEO satellites are often used as telecommunication satellites, allowing antennas on the Earth to remain at one location. This effect is strengthened by the effect that one satellite can service a significant portion of the Earth's surface, needing only three satellites to achieve nearly global coverage. The distance from which these satellites orbit is what makes them useful for certain telecommunications and as weather satellites, but this can also be a drawback, due to the enormous distance between a satellite and a user, communications can be slow. [3]

MEO satellites can cover a large range of altitudes and occupy the space between LEO and GEO. MEO satellites are largely used for navigation such as GPS on your phone. One example of this is the Galileo constellation. Again these satellites can also suffer from slow communications due to long round trips the data has to make. [3]

1.3 Comparison with Terrestrial Networks

1.4 Problem Statement

Despite investment in Ireland's terrestrial broadband infrastructure through initiatives such as the National Broadband Plan [4], substantial challenges remain in delivering consistent high-preforming broadband connectivity to rural and sparsely populated ares. The Economic and technical limitations with fibre deployment and mobile network density results in long rollout times with uneven service availability. While LEO constellations have emerged as a potential alternative capable of providing coverage for large areas with reduced dependency on ground based infrastructure, their practical suitability for

addressing Ireland's rural connectivity challenges should be sufficiently quantified to measure how much these constellations could either provide their own service or complement an existing network.

1.5 Research Objectives

The key objective of this project is to evaluate if LEO satellite constellations are capable of providing a comparable or improved broadband service to 5G across rural Ireland which may be under-served under the NBP [1]. A secondary goal is to assess whether LEO systems are suitable to be part of a complementary solution to existing broadband networks. In order to achieve this a simulation framework for modelling must be developed which is capable of estimate the theoretical capacity achievable by LEO satellite links.

1.6 Scope of Project

The scope of this project is focused on evaluating the feasibility of LEO satellite constellations as a solution for rural broadband connectivity in Ireland. The work concentrates on modelling and analyzing satellite coverage availability and theoretical communication performance using a simulation based approached with publicly available data.

This project includes the development of a software simulation for LEO satellites using real Two-Line Element (TLE) data. The simulation will evaluate satellite visibility, coverage, and theoretical data rates across a grid of locations in Ireland. The project will compare the performance of LEO satellite systems against terrestrial broadband solutions in terms of coverage and potential data rates.

The geographic scope is limited to Ireland, with a particular emphasis on rural areas that are currently under-served by terrestrial broadband. The project will focus on existing and planned LEO constellations, such as Starlink, OneWeb, and Kuiper, to assess their potential impact on rural connectivity in Ireland.

Several aspects are intentionally excluded from the scope of this project. The simulation does not account for detail processes such as modulation schemes, encoding, adaptive power control or access scheduling. Atmospheric effects, terrain obstructions, attenuation from weather, and traffic from user demand are not explicitly considered. On top of this, network level behaviors such as gateway placement, routing strategies, handover delays, and are not within the scope of this project.

By constraining the scope, this project aims to deliver a clear, reproducible, any computationally achievable assessment of whether LEO satellite systems can meaningfully enhance broadband connectivity in rural Ireland.

1.7 Thesis Structure

This thesis is organized into six chapters.

Chapter 1 introduces the motivation for the study and outlines the challenges of broadband connectivity in rural Ireland, and the potential of LEO satellite systems as a solution. It presents the problem context, research objectives, scope of the work and the structure of the thesis.

Chapter 2 reviews literature on LEO satellite communication systems, rural broadband connectivity, non-terrestrial networks, and LEO coverage in other countries.

Chapter 3 describes the system design and methodology used in the this project. This chapter details the modelling approach, data sources, orbit propagation, coordinate systems, and the assumptions which are used throughout the project.

Chapter 4 describes the software implementation of the project. The simulation architecture, data processing workflow, coordinate transformation logic, visibility and coverage checks, performance metrics, optimization of the project and computational complexity are all discussed in this chapter.

Chapter 5 presents the results of the simulations, including coverage maps, availability statistics, and theoretical throughput estimates for the evaluated LEO constellations, and comparative performance against terrestrial broadband solutions.

Finally, Chapter 6 concludes the thesis by summarizing the key findings, discussing their implications for rural broadband connectivity, and suggesting directions for future research in this area. This chapter will also reflect on the limitations of the study and potential improvements to the simulation model.

2. Literature Review

2.1 LEO Satellite Constellation Architectures

Compared to geostationary satellites, LEO constellations offer significantly lower propagation delay and reduced path loss, enabling a high quality broadband service [5] [6]. With LEO continuous service is achieved through coordinated constellation geometry and handover logic between satellites as opposed to fixed satellite positioning.

LEO constellations are generally organized into multiple orbital planes, with satellites even spaced within each plane to ensure consistent regional coverage. Key architectural parameter include orbital altitude, inclination, number of orbital planes, and the number of satellites per plane. As the orbital altitude decreases, the latency reduces, however a larger total satellite count is then required to account for coverage gaps [5] . These design trade-offs include system cost, performance and scalability.

LEO satellite constellation architecture is a complex trade-off between coverage, latency, and system complexity. Understanding the cost and result of key design features is crucial for successful LEO constellation deployment.

2.2 Existing LEO Constellations

Table 2.1: Common frequency ranges for Ku- and Ka-band satellite communications [7].

Band	Frequency (GHz)
Ku-band	12 – 18
Ka-band	18 – 26

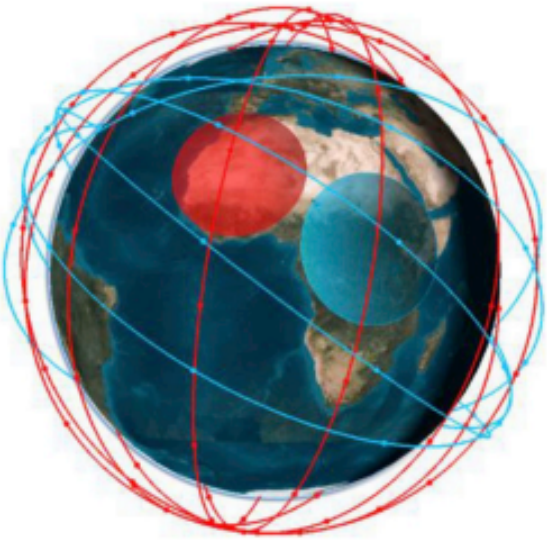
In recent years, several companies have begun deploying extensive constellation in order to achieve global broadband services. The following section will give an overview of some of these companies.

Telesat

Telesat's LEO constellation operates in the KA-band and is designed to provide global broadband coverage, with high throughput and low latency. Telesat's constellation uses 1.8GHz of the KA-band with a downlink (17.8-20.2GHz) and an uplink (27.7-30.0GHz) stream, and is made up of two sets of orbits. This consists of, six polar orbits at an altitude of Ĩ000Km and 5 inclined orbits at Ĩ200km. These orbits contain at least 12 and 5 satellites, respectively. The polar orbits are focuses on providing full global coverage while the inclined orbits mainly focus on the middle latitudes where most of Telesat's customers are.

In both of these sets of orbits, optical inter-satellite links are employed to ensure that continuous connectivity is secured. Each satellite in the constellation can functions as an IP node and has an on board digital signal processing module supporting demodulation, routing and remodulation.

Each satellite uses a direct radiation array which means it can create multiple narrow communication beans instead of a singular broad signal. These beams can be steers ensuring that the satellite can ensure coverage to various locations on the ground when user location or demand changes. This ensures the system is power efficient, not wasting bandwidth and energy in areas where there are no users. [8]



(a) Telesat Constellation [8]

(b) OneWeb Constellation [8]

Figure 2.1: Map of Telesat and OneWeb Constellations

2.2.1 OneWeb

Unlike Telesat, OneWeb operates in the Ku and Ka bands and contains 720 satellites in 18 circular orbits orbitting at Ĩ200Km. Each satellite operates on a 'bent pipe payload' system, meaning signals received from users on the ground are amplified and forwarded to a gateway instead of being routed on board the satellite itself. Each satellite generates 16 fixed beams to ensure that users are always in view of at least one satellite.

While the Ku-band is dedicated to users, the Ka-band is dedicated to gateway links. Each satellite has two steerable antennas, one in operation and a backup antenna. Unlike many other constellations, no satellites in OneWeb's constellation uses inter-satellite links, meaning connectivity is solely reliant on whether the user and gateway are in line of sight with a satellite. [8]

2.2.2 Starlink

SpaceX's Starlink also operates in the Ku and Ka bands and is planned to contain several thousand satellites in multiple types of orbits. Initial deployment consisted of 1600 satellites orbiting at 550Km. Each satellite has a phased array communications payload allowing it to generate multiple steerable beams which can be adjusted depending on the location of user demand and can deliver a latency of only 25ms. [9]. [8]

Starlink also deploys optical inter-satellite links.This allows satellites to communicate with one another, meaning data can be routed across the constellation without relying on ground base stations or gateways which improves coverage in remote locations. Much like OneWeb, the Ku band services the user links while the ka band services the gateway links. [8]

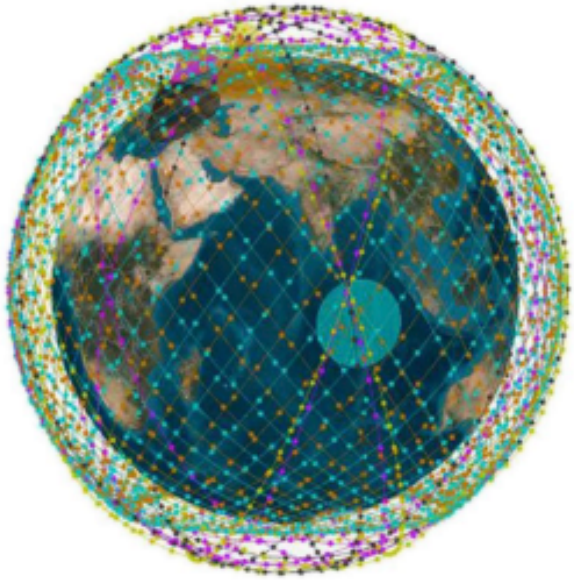


Figure 2.2: Starlink constellation [8]

2.2.3 Summary

Table 2.2: Comparison of LEO broadband satellite constellations.

System	Bands	Satellites	Altitude (km)	Inclination (°)	Sat./Plane	Payload	ISL
Telesat	Ka	≥117	1000 / 1200	99.5 / 37.4	≥12 / ≥10	Digital processing (DRA)	Yes
OneWeb	Ku/Ka	720	1200	87	40	Bent-pipe (fixed beams)	No
Starlink (Phase 1)	Ku/Ka	1600 (→4425)	550	53–81	50–75	Phased-array (steerable)	Yes

Table 2.2 highlights some of the key design strategies among existing LEO constellations. Starlink is unique with the largest constellation and lowest orbit among this group of constellations. This enables Starlink's low latency world-wide coverage. Telesat with fewer satellites but advances processing of payloads utilizes inter-satellite links for efficient data routing, while OneWeb using a bent pipe architecture without ISLs, making it much more dependant on ground gateways. Overall, satellites systems with ISLs and dynamic beam forming have greater flexibility and are superior for rural connectivity compared to OneWeb's traditional approach which is reliant on terrestrial infrastructure.

2.3 Coverage and Latency Characteristics of LEO Systems

2.4 Capacity and Throughput Modelling in Satellite Networks

2.5 Challenges and Limitations of LEO Systems

Despite the rapid growth of LEO constellations, there are technical and regulatory challenges. It is crucial that these issues are understood in light of this project, particularly in the context of Direct-to-Cell technology.

2.5.1 Orbital congestion and Space debris

The deployment of thousands of satellites introduces risk of orbital congestion and collisions. Congestion in low Earth orbit reached record highs in 2023, with over 2800 satellites entering LEO during the year. The majority of these satellites in LEO are commercial communications constellations. On top of this, there are over 35,000 objects in orbit that are tracked by space surveillance networks. 26,000 of these are larger than 10cm and a further one million pieces are larger than 1cm. These have the potential to cause serious issues for operating satellites and force operators to spend limited fuel resources to preform maneuvers. In 2023 the ESA created the Zero debris charter to encourage others to become debris neutral by 2030 [10].

2.5.2 Spectrum and Regulations

Spectrum allocation and regulations are some of the most significant challenges for LEO broadband systems. Unlike terrestrial networks which operation within defined know frequency bands, LEO constellations require global coverage to avoid interference with existing services. While most LEO systems operate within the KU and KA band (12-18Ghz and

18-26Ghz) [7], these frequencies are shared with other satellite and terrestrial services. This creates conflicts which have to managed through international agreements and licensing.

The International Telecommunication Union (ITU) governs the global spectrum allocation while national regulatory services such as ComReg in Ireland enforce local compliance. Any operator must have a license for uplink and downlink frequencies, ensuring compatibility with existing networks. Due to the growing number of LEO constellations, complexity in frequency coordination has increased, and this requires interference mitigation techniques such as dynamic beam forming and adaptive power control [11].

Regulation frameworks are changing alongside this technology, 3GPP standards now enable the integration of satellite and terrestrial networks [12]. However challenges still remain in areas such as managing congestion of busy frequency bands. Should these issues not be addressed issues with service degradation, legal disputes, and delays in deployment could begin to take place.

2.6 LEO Satellite Coverage Analysis in Other Countries

Several countries with with large rural or remote populations have adopted LEO satellite communications to provide broadband access where terrestrial networks are economically feasible.

2.6.1 Canada

In Canada, LEO satellites have been used to address broadband deficiencies, in northern and remote regions. Both Starlink and OneWeb have been deployed to provide last-mile connectivity or satellite back-haul to local wireless networks. Government-backed initiatives have supported the integration of LEO services to complement terrestrial infrastructures, recognizing how they can lower latency and higher capacity over traditional satellite and terrestrial links. Canadian studies show that LEO systems offer a cost effective alternative [13].

2.6.2 Australia

Australia has formally integrated LEO broadband into its national broadband strategy. In 2025, the government owned National Broadband Network (NBN) annoyed a transition from geostationary satellites to LEO satellites for regional and remote users in partnership with Amazons Kuiper. The goal of this is to provide a lower latency and higher quality broadband to rural and remote premises number over 300,000. This use case shows LEO satellites can provide broadband performance close to terrestrial services, particularly in rural areas [14].

2.6.3 United States of America - Alaska

Alaska has one of the most well developed deployments of LEO satellite broadband. Due to the rugged geography, low population density and permafrost conditions, fibre deployment is exceptionally expensive and challenging in many regions. As a result, large areas of rural Alaska have historically relied on geostationary satellite systems or low-capacity legacy connections, which offer limited performance and high latency.

In recent years, LEo satellite systems - primarily Starlink - have been adopted across rural Alaskan communities. These services are being used by households as well as public institutes such as schools, healthcare facilities, and emergency services. Deployment reports show that LEO broadband has enabled reliable internet, with government funded programmes which incorporate LEO services for undeserved regions [15].

Independent performance measurement analyses further shows the benefits of LEO systems in Alaska. COmpared to traditional Geostationary services, LEO connectivity has been shown to significantly reduce latency and improve overall service reliability. LEO broadband now form a core component of Alaska's broadband strategies, with government funded programmes incorporating LEO services. [16]

Alaska is set to receive $1.2 billion in funding from the federal government's Broadband Equity Access and Deployment (BEAD) program, of which $629 million has been set aside for broadband projects. This will be used to connect 44,734 under-served or unserved locations across Alaska. Due to the previously discussed geography issues, this initiative will use a mix of technologies. 14,400 of these locations will be served with Starlink. A breakdown of how this broadband will be rolled out can be seen in Figure 2.3.

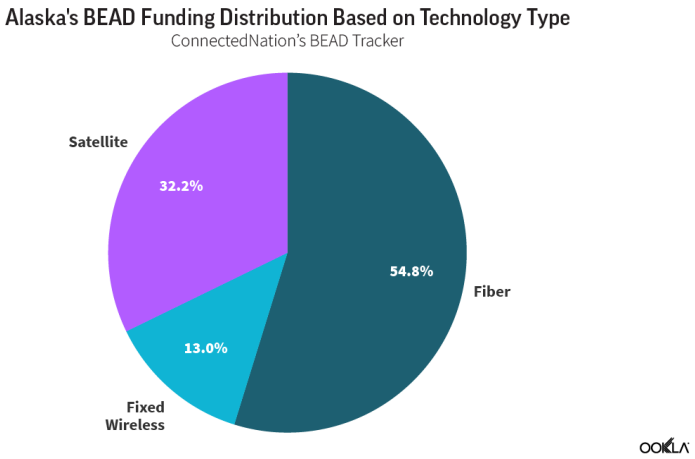


Figure 2.3: Alaska's BEAD Funding and Distribution Based on Technology Type [16]

Starlink in 2025 was able to bring up the percentage of Ookla speedtest users in Alaska with broadband speeds of 100/20 Mbps from 4.35% in the first half of 2025 to 15.9% in the second half of 2025. In fact a greater number of rural Starlink users in Alaska are receiving broadband speeds of 100/20 Mbps than it's urban users. By the end of 2025 16.8% of Starlink users in rural areas were able to get broadband speeds of 100/20 Mbps compared to just 12.3% of urban users [16].

According to Ookla, Starlink's median download speeds rose from 73.5 Mbps in March 2025 to 108.5 Mbps in March 2026, and it's median upload speeds go from 10.7 Mbps in March 2025 to 17.1 Mbps in March 2026. This increase in speed can be seen in Figures 2.4 and 2.5.

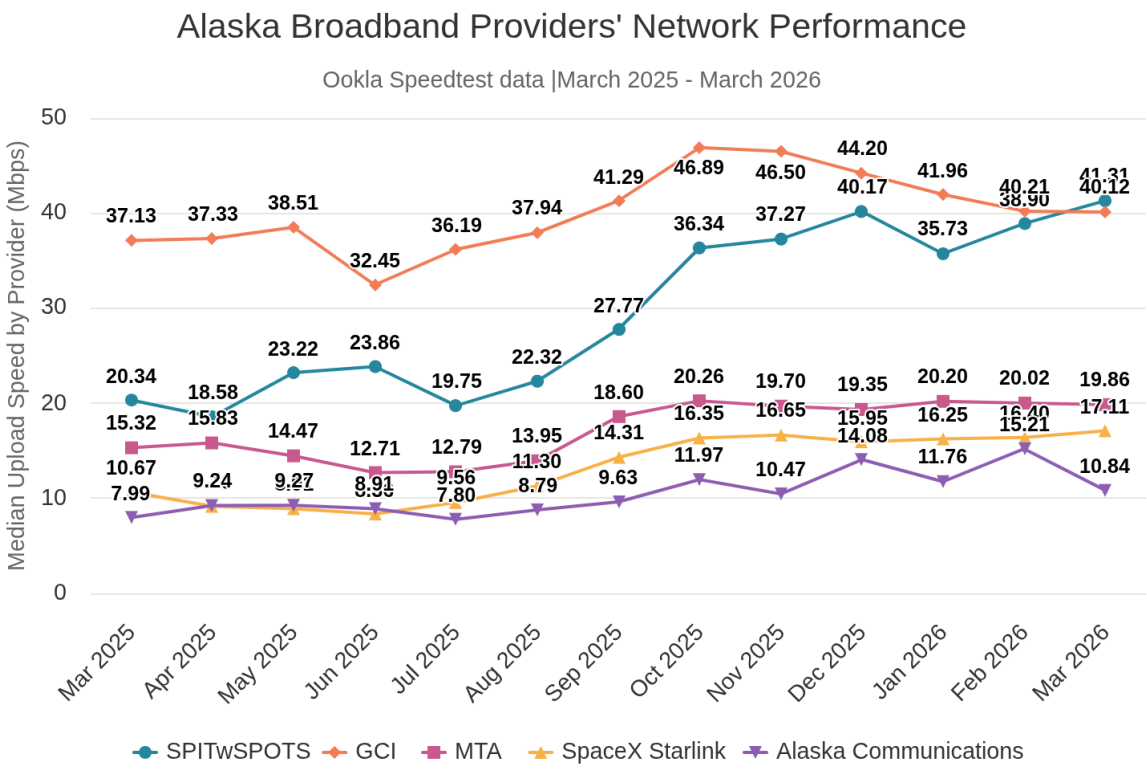


Figure 2.4: Alaska broadband plot of downlink speeds.

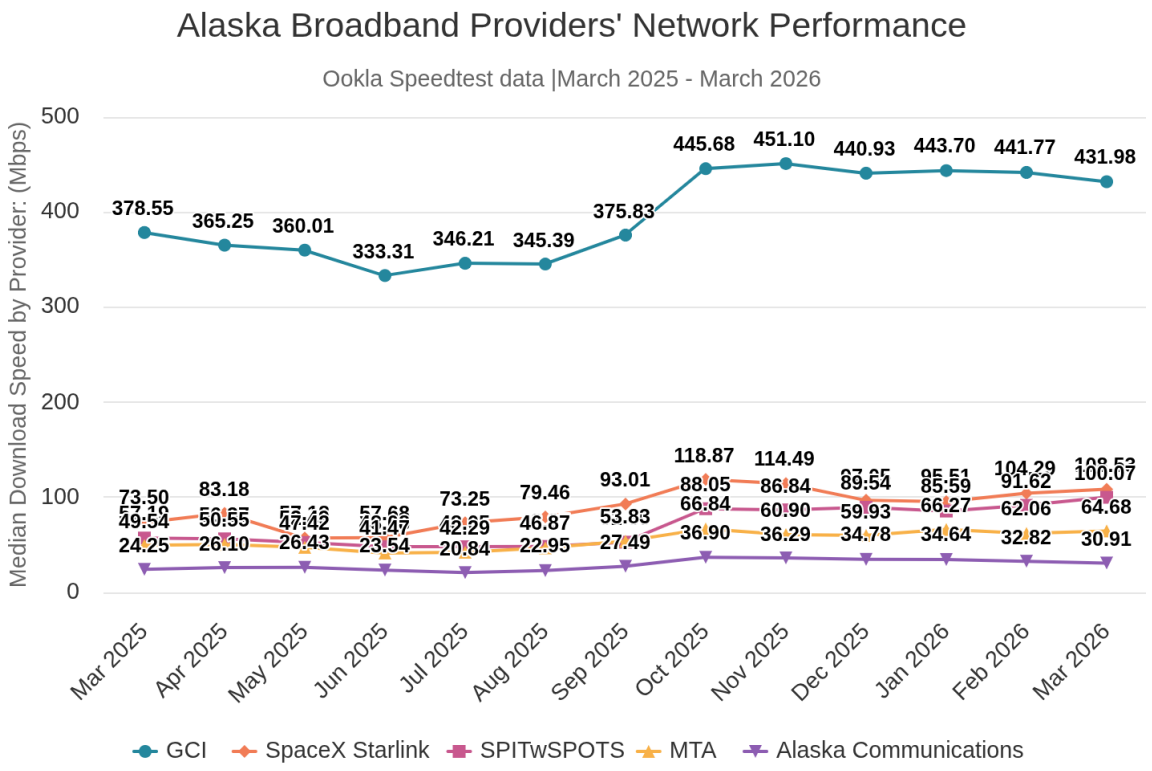


Figure 2.5: Alaska broadband plot of uplink speeds.

This provides solid evidence of a real world example of how LEO satellite broadband can function as along term connectivity solution. While there are differences in Ireland such as environmental differences, the underlying challenges are the same - low population density, high infrastructure costs, and limited commercial incentives for fibre deployment. Alaska is a compelling case to further investigate the potential for LEO broadband in Ireland.

2.6.4 Relevance to Ireland

These international examples demonstrate that LEO constellations are increasingly becoming a viable option for providing broadband in rural areas. Common across all cases are low population density, and challenging geography with high costs for terrestrial network deployment. These conditions align closely with what can be found in regions in Ireland.

2.7 LEO Satellites and Direct-to-Device

Direct to Device (D2D) satellite communication is a new approach to global connectivity by enabling LEO satellites to communicate directly with user's devices such as smartphones without relying on intermediate ground stations or gateways. This addresses one of the main challenges in telecommunication networks, the cost of terrestrial infrastructure in rural areas. Rural areas can suffer from poor broadband connectivity due to the high cost of fibre optic and cellular towers. Traditional infrastructure requires significant investment, which is rarely justified by the low subscriber density in these areas. D2D is not held back by these limitations by not needing anything other than existing consumer devices. This completely eliminates the need for specialized terminals and cell towers. This is a viable solution for rural Irish communities where people remain under-served.

D2D communication enables satellites to communicate directly with an end user device without needing any sort of intermediate infrastructure. There are three main approaches, the Mobile satellite service spectrum (MSS), which is traditionally reserved for satellite services, and the LMS (Location and monitoring service) and IMT (international mobile telecommunications) [17] Recent developments in 3GPP have formalized the NTN specifications, making it possible to integrate satellite connectivity into terrestrial mobile networks seamlessly.

However, despite these limitations in recent years some companies have began pursuing D2D solution. In particular starlink and T-Mobile have partnered to provide coverage in dead zones where no coverage previously existed. Their focus is largely on rural and remote areas. As satellite constellations expand D2D is set to become a major part of broadband. It has the ability to complement terrestrial networks and provide a reliable connection during even natural disasters. [18]

2.7.1 Upgrading Existing Satellites for D2D Capability

While there are some D2D satellite services being developed in new launces of satellites, there is also an interest in enabling D2D functionality on existing LEO constellations which were previously not capable of this. Retrofitting or incrementally upgrading satellites could represent a cost effective path to accelerate D2D deployment.

One method of upgrading for D2D is software-defined payload architectures. Many modern LEO satellites are already contain flexible, reprogrammable radio hardware that allows waveform and protocol updates through software. This enables satellites which were originally designed for fixed broadband to support new bandwidths compatible with territorial mobile networks such as LTE or 5g NTN, without any hardware replacement.

3. System Design and Methodology

3.1 Data Inputs

The main data input to the propagation of satellites in this project is Two-Line Element sets (TLEs). TLEs are a common publicly accessible format which describes Earth-orbiting satellite positions. The primary source of TLE data is SpaceTrack [19]. Although SpaceTrack is the primary source, many projects such as this project and other research rely on sources such as CelesTrak, which takes the data from SpaceTrack and organizes it into specific groupings [20]. Some examples of these groupings are shown in Listing 3.1.

- **Starlink**
- **ISS**
- **Earth observation satellites**
- **Weather satellites**
- **Communications satellites**
- **Active vs. debris groupings**

Figure 3.1: Common satellite groupings provided by CelesTrak

While other sites exist such as N2Y0 and AMSAT, CelesTrak remains one of the most reliable and widely used platforms due to consistent updates synchronized with SpaceTrack and curated categories which help simplify data access. CelesTrak updates the majority of its TLE files approximately every two hours. Its Starlink data can expect multiple updates per day, making it suitable for this project's application [21].

3.2 Two Line Element Data

TLEs are a standardized format used to describe the orbital properties of Earth-orbiting satellites. A TLE represents a satellite's orbit at a specific moment in time, known as the epoch. Because the TLE format is simple and lightweight, it is widely supported and remains the most common method for distributing orbit data for tracking and prediction.

A TLE consists of three lines: line 0 (optional) contains the satellite name, line 1 contains identification and basic orbital parameters, and line 2 describes remaining orbital elements such as inclination. This is summarized in Table 3.1. An example of TLE data is shown in Listing 3.2.

```
STARLINK-5694
1 55481U 23015AJ  26041.24424508  .00002920  00000+0  23114-3 0  9992
2 55481  42.9978 128.2652 0001315 248.0292 112.0411 15.02536911166474
```

Figure 3.2: Two-Line Element (TLE) for STARLINK-5694

Table 3.1: Summary of key parameters in a Two-Line Element

Field	**Value**	**Meaning**
Satellite Name	STARLINK-5694	Human-readable satellite identifier
Catalog Number	55481	Unique NORAD satellite ID
Classification	U	Unclassified satellite
Launch Year	2023	Part of the International Designator (23)
Launch Number	015	15th launch of the year 2023
Launch Piece	AJ	Specific object from that launch
Epoch (Year)	2026	TLE epoch year (26)
Epoch (Day)	41.24424508	Day of year when the TLE is valid
Inclination (deg)	42.9978	Tilt of orbit relative to Earth's equator
Mean Motion (rev/day)	15.02536911	Orbits per day (≈95.8 min period)

3.3 Orbit Propagation

TLE data describes a satellite's orbit at a specific moment in time, know as the epoch. To determine a given satellite's position at times throughout the simulation, an orbit propagation model is required. Orbit propagation uses the orbital parameters contained within the TLE along with a model for orbital perturbations to predict the satellites's future position and velocity as a given time stamp

In this project, the Simplified General Perturbations Model 4 (SGP4) [22] is used for orbit propagation. SPG4 is a widely used algorithm for propagating satellite orbits for organizations such as NORAD and NASA [23]. THe model accounts for dominant perturbations which affect LEO satellites such as Earth's uneven gravitational field, and atmospheric drag. SGP4 is computationally efficient and provides sufficient accuracy for coverage estimation.

For each satellite, SGP4 computes a time varying state vector consisting of position and velocity in the True Equator Mean Equinox (TEME) frame. This allows the system to determine where each satellite is located at any given time during the simulation, which is essential for evaluating visibility and coverage. These are calculated at discrete time intervals defined by the simulation parameters, such as every 10 seconds, to balance accuracy and computational load.

3.4 Coordinate Systems

TO determine satellite visibility and coverage, satellite orbits and ground user locations must be expressed in a common coordinate system. This allows for accurate calculations of line-of-sight and beam-width constraints. Different coordinate systems are used at different stages of the analysis, with each serving a distinct purpose in representing satellite motion or Earth fixed locations. This sections aims to introduce the coordinate systems used in this project and their roles in the methodology. The mathematical transformations between these coordinate systems will be discussed in section 4.4.

In this project, the satellites which are propagated using SGP4 are expressed in the True Equator Mean Equinox (TEME) frame, while ground sites are typically given in geodetic coordinates. It is therefore necessary to convert both to an Earth-Centered Earth-Fixed (ECEF) frame [24, 25]. This provides a common reference frame for enable geometric calculations such as line-of-sight and beam-width evaluations.

3.4.1 TEME

TEME is the native output frame of SGP4. It is non-rotating with respect to Earth's surface and must be converted before comparison with geodetic locations. TEME is an Earth centered reference frame whose orientation is relative to Earth's equatorial plane and mean equinox. It is designed specifically for use with TLE data and the mathematical approximations which are employed in SGP4 [24].

TEME is well suited for representing satellite motion because it provides a stable reference for orbit prediction that is not fixed to the Earth's surface which rotates. However, as this coordinate system does not rotate with the Earth, positions expressed in this frame cannot be directly compared with ground locations which rotate with the Earth directly.

3.4.2 ECEF

ECEF is a rotation cartesian coordinate system which is fixed relative to the Earth's surface. It's origin is located at the Earth's center of mass, with axes alinged to the Earth's rotation and geographic geometry. As it rotates with the Earth, ECEF is ideal for representing ground locations. ECEF is used to describe the observer's positions which come from latitude, longitude coordinates, and provides a common reference frame for evaluation relationships between satellites and ground locations. By expressing both the ground user and the satellite in ECEF, direct calculations such as line-of-sight and beam-width evaluations can be carried out [26]. a Comparison between geodetic and ECEF coordinates is shown in Figure 3.3

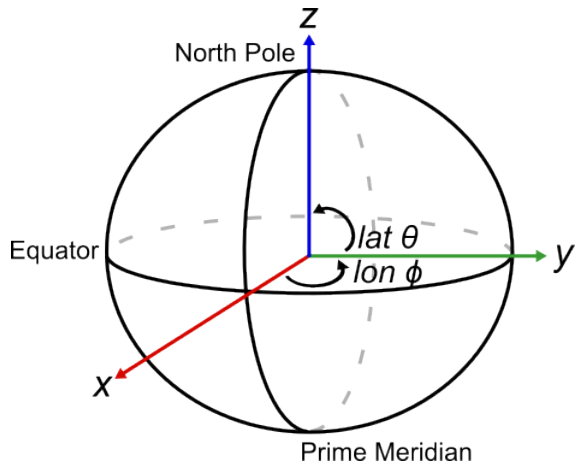


Figure 3.3: ECEF Coordinate System [27]

3.5 Assumptions

To simplify the analysis and focus on the core aspects of satellite coverage, several assumptions are made in this project. These assumptions simplify the modelling process while preserving sufficient accuracy for the intended analysis.

The first assumption is that the satellites are assumed to follow trajectories predicted by SGP4 when initialized with publicly available TLE data. While SPG4 accounts for things such as Earth's oblate shape and atmospheric drag, it does not capture all short term maneuvering. Satellites are assumed to follow the predicted trajectories without performing maneuvers such as station keeping or collision avoidance.

The second assumption is that the Earth is modelled using a WGS-84 ellipsoid for al geodetic to ECEF transformations. Local terrain variation, orography, and surface irregularities are not considered. Ground users are assumed to be at sea level unless otherwise specified, visibility issues due to terrain and other obstructions such as buildings or vegetation are not considered. A free space path model loss model is instead implemented. Even with this free space path loss model, atmospheric effects such as rain, wind and any other weather related fading effects are not included in the analysis. There is a constant loss factor per unit of distance which is assumed.

The third assumption is the antenna radiation patterns are simplified using a fixed half-power beam width model. Each satellite is assumed to have a single, ideal beam centered on its nadir direction, and beam forming, side-lobe effects, and other complexities of real antenna patterns are not considered. The beam width is assumed to be constant and does not vary with frequency or other factors. Beam-t0-ground aligned is evaluated purely geometrically, based on angular separation without accounting for adaptive beam scheduling or multi-beam operation. To attempt to account for this a relatively wide beam width of 15 degrees is used, which is wider than the typical beam width of a LEO satellite, to provide a more conservative estimate of coverage.

The fourth assumption is that interference between satellites is modelled in a simplified manner. If multiple satellites are visible to a ground user at the same time, it is assumed that the satellite with the strongest signal (i.e., the smallest angular separation from the user's zenith) provides coverage, while others do not. Interference power is estimated by summing the reduced contributions of all visible satellites transmit power power based on a constant scalar.

The sixth assumption is that handover dynamics are not explicitly modelled. WHen multiple satellites satisfy coverage criteria, the satellite which provides service is assumed to be the one with the best geometric alignment. Latency associated with handover, beam switching, and gateway routing is neglected.

The seventh assumption is that user demand is assumed to be uniform across the coverage grid. Traffic load, contention between user, and scheduling policies, are outside the scope of this project. Throughput is modelled using the Shannon capacity formula to represent theoretical data rates.

Finally, the analysis assumes that satellites are operate continuously and are always transmitting at their maximum power when they are visible. Satellite maintenance periods, power constraints, and operational outages are not considered.

4. Implementation

This chapter describes the logic behind the software implementation of the satellite coverage and capacity analysis which is detailed in Chapter 3. The system is implemented in Python and is designed to evaluate the availability and theoretical throughput of LEO satellite constellations for rural regions in Ireland.

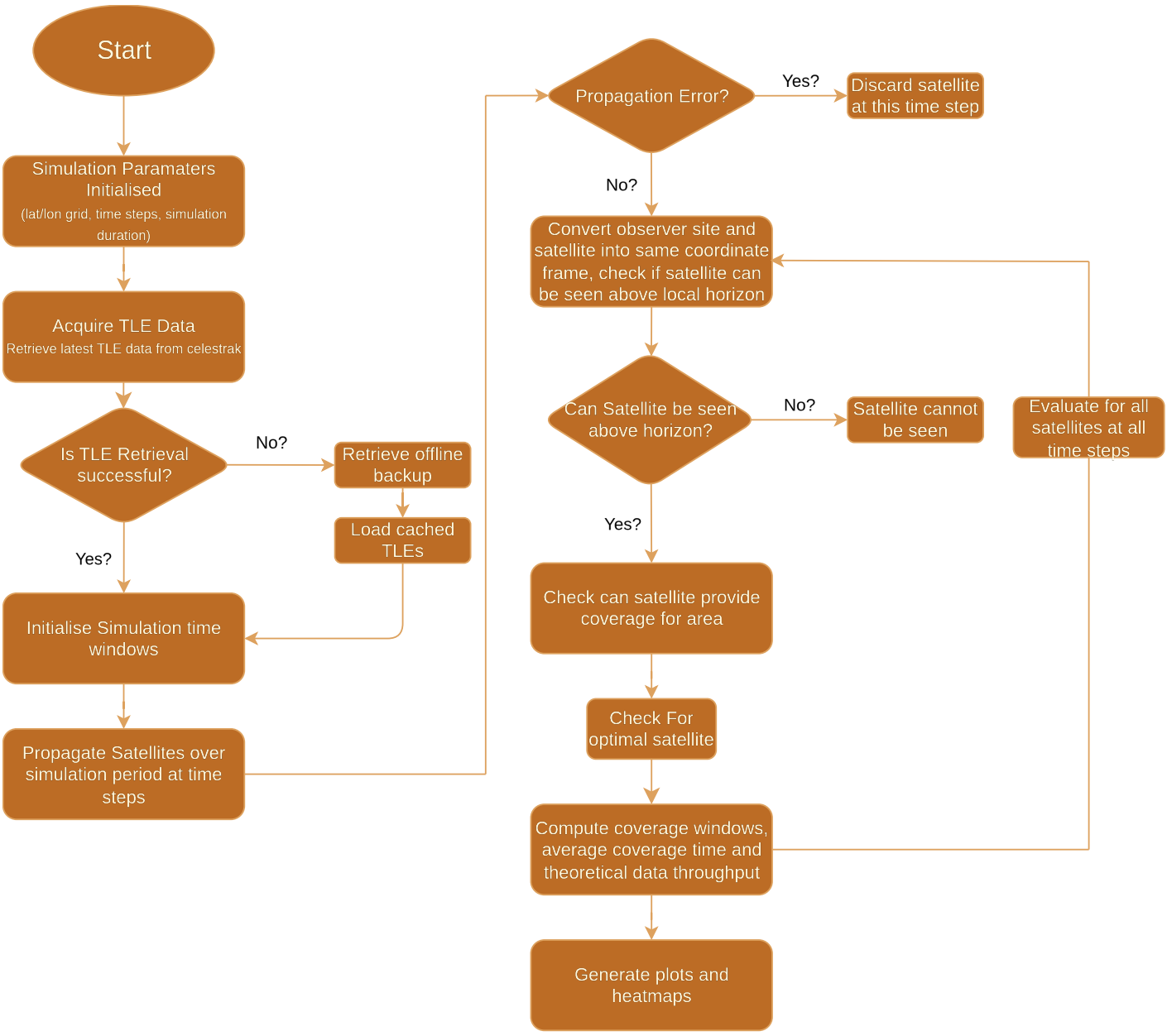


Figure 4.1: Implementation Logic

Figure 4.1 illustrates the overall execution and flow of the implementation logic. The implementation begins by setting the simulation Parameters, i.e, duration of simulation, frequency of time steps, the density of the coverage grid, and the size of the area being studied. Following this, the TLE data is then acquired from CelesTrak, which is used to propagate satellite orbits using the SGP4 model. At each discrete time step, satellite positions are propagated and then transformed into a common Earth fixed coordinate frame, and evaluated using an elevation mask and beam width constraints to determine which satellites can provide coverage and when.

The core execution of this system is based in a loop which iterates over a grid of latitude and longitude coordinated within which visibility and coverage check are preformed for all propagated satellites at each time step. After the completion of this simulation, the collected data is stored and then processed into heatmaps to be visualized, this data will then be compared tio publicly available terrestrial data.

4.1 Simulation Parameter Initialization

The first stage of implementation involves defining a set of simulation parameters which will be utilized in the later stages of implementation. The core parameters for the simulation are as follows:

- **Simulation Duration**: The total time span for which the simulation will run, in hours.
- **Simulation Start Time**: The UTC timestamp marking the beginning of the simulation period.
- **Time Step**: The interval at which satellite positions are propagated and visibility checks are performed, in seconds.
- **Grid Resolution**: The spacing between latitude and longitude points in the coverage grid, in degrees.
- **Area of Interest**: The latitude and longitude ranges of the area over which coverage will be evaluated.
- **Beam Width**: The half power beam width of the satellite's antenna, in degrees, which defines the coverage area of the satellite.
- **Elevation Mask**: The minimum elevation angle above the horizon required for a satellite to be considered visible, in degrees.
- **Bandwidth**: The communication bandwidth available for the satellite link, in MHz.
- **Noise Figure**: The noise figure of the receiver, in dB, which quantifies the degradation of the signal-to-noise ratio due to the receiver's internal noise.
- **Transmit Power**: The power at which the satellite transmits signals, in dBW.
- **Receiver Gain**: The gain of the receiver's antenna, in dBi, which quantifies its ability to capture incoming signals.
- **Frequency**: The frequency at which the satellite operates, in GHz, which affects the path loss and atmospheric attenuation.
- **TLE Data Source**: The source from which TLE data will be acquired, such as CelesTrak, and any fallback options for offline data.

4.2 Data acquisition

The next stage of the implementation is acquiring up-to-date orbital Parameters for the given LEO constellation(Starlink, Kuiper, OneWeb). TLE sets are retrieved from Celestrak. This data provides the perturbations required to propagate the orbits using SGP4. The TLE data used in this system are taken directly from Celestrak [20]. Downloading this data produces a text stream which has sets of three lines of text blocks, satellite name, followed by line 1 and line 2 of the TLE. The implementation of this system parses these entries for each satellite.

Using the most up to date TLEs ensures that the system accurately represents the dynamic nature of large LEO constellations, which orbital parameters may change drastically and regularly due to factors such as manual maneuvers.

Acquiring TLE data can be done through a link from CelesTrak. This project downloads raw TLE from CelesTrak using an HTTP request and then parses the response into a list of lines for processing. Python makes this simple with a requests library, which can be seen implemented in listing 4.1

Listing 4.1: Pulling TLE data

```
response = requests.get("https://celestrak.org/NORAD/elements/gp.php?GROUP=starlink&
    FORMAT=tle")
lines = response.text.splitlines()
```

However, this can create issues, as this project is pulling TLE data from thousands of satellites over many time steps, this can create a large number of HTTP requests which can be blocked by CelesTrak. To avoid this, the TLE data is pulled once at the start of the simulation and then stored locally for the duration of the simulation. This allows for a more efficient and robust implementation, as it avoids potential issues with rate limiting and ensures that the same TLE data is used throughout the simulation for consistency.

On top of this it is critical to ensure that the TLE data is parsed at the right location in the code, as seen in Figure 4.1, this project employs a 0.1 lat lon step for mapping over Ireland, which equates to 1000s of grid positions, and over a duration of 24 hours with a time step of 10 seconds, in order to prevent temporary IP bans, the TLE data must be pulled first, then stored, then used.

Even with this approach, this project experienced some rate limiting, throttling and temporary IP bans from CelesTrak, to get around this, as offline backup of the TLE data was stored and manually updated periodically. This can be seen implemented in Listing 4.2. Should there be issues with accessing data directly from CelesTrak, it was still possible to, design and test while using an offline backup file containing TLE data.

Listing 4.2: TLE data acquisition logic

```
        try:
        if source.get("type") == "online":
            response = requests.get(source["url"], timeout=8)
            response.raise_for_status()
            lines = response.text.strip().splitlines()
        else:
            raise Exception
    except Exception:
        here = os.path.dirname(os.path.abspath(__file__))
        path = os.path.join(here, "..", source["local"])
        with open(path, "r") as f:
            lines = f.read().strip().splitlines()

    return lines
```

It is important to note that while this approach ensures consistency and avoids rate limiting, it may not capture any orbital changes that occur during the simulation period. For long simulations, spanning multiple days, it may be necessary to implement a strategy for periodically updating the TLE data to ensure that the simulation remains accurate over time.

Once TLE data is acquired, it is parsed and stored in a structured format for use in the SGP4 propagation stage. Each satellite's TLE is associated with its name and catalog number, allowing for easy reference during propagation and visibility analysis. There are two types of TLE data, the first is seen before in Listing 3.2 , with lines 0, 1, and 2, the second type of TLE data does not include line 0, and only has lines 1 and 2 and example of this is shown in Listing 4.3, which is a OneWeb satellite.

Listing 4.3: Two-Line Element (TLE) for SATELLITE-44057

```
1 44057U 19010A   26102.43981189  .00000080  00000+0  17451-3 0  9993
2 44057  87.9055 241.8469 0002002 112.4154 247.7189 13.16593523342917
```

The implementation of this project accounts for both formats, ensuring that all available TLE data can be utilized effectively, and can be seen in 4.4

Listing 4.4: TLE data acquisition logic

```
    tles = []
    i = 0

    while i < len(lines) - 1:

        # Case 1: bare 2-line TLE
        if lines[i].startswith("1 ") and lines[i + 1].startswith("2 "):
            line1 = lines[i].strip()
            line2 = lines[i + 1].strip()
            name = f"SAT-{line1[2:7]}"
            i += 2
        else:
            name = lines[i].strip()
            line1 = lines[i + 1].strip()
            line2 = lines[i + 2].strip()
            i += 3

        tles.append((name, line1, line2))

    return tles
```

This results in an array which can be easily iterated over for propagation and visibility checks, an example of this array is shown in Listing 4.5.

Listing 4.5: In-memory representation of parsed TLE data as stored in the `tles` array

```
tles = [
    (
        "SAT-44057",
        "1 44057U 19010A   26102.43981189  .00000080  00000+0  17451-3 0  9993",
        "2 44057  87.9055 241.8469 0002002 112.4154 247.7189 13.16593523342917"
    ),
    (
        "STARLINK-5694",
        "1 55481U 23015AJ  26041.24424508  .00002920  00000+0  23114-3 0  9992",
        "2 55481  42.9978 128.2652 0001315 248.0292 112.0411 15.02536911166474"
    )
]
```

4.3 Orbit propagation

Once the TLEs have been retrieved and parsed in a suitable manner, each satellite's position and velocity are computed using the SGP4 algorithm. In this system each satellites state vector can be generated as seen in Listing 4.6.

Listing 4.6: SGP4 propagation example

```
satellite = Satrec.twoline2rv(line1, line2)
e, r, v = satellite.sgp4(jd, fr)
```

The outputs of the 4.6 are as follows:

- **r**: The satellite position vector (x, y, z) expressed in the True Equator Mean Equinox (TEME) reference frame. This can be described as the satellite's position in respect to Earth's center of mass.
- **v**: The satellite velocity vector in the TEME coordinate frame. Also in respect to Earth's center of mass, this vector describes the satellite's velocity in three-dimensional space.
- **e**: An error flag returned by the propagator, where a value of zero indicates successful propagation and non-zero values correspond to specific SGP4 error conditions.

The Julian Date (jd, fr) is computed using the current UTC timestamp to ensure a precise propagation epoch. Any satellite with a non zero error flags are discarded to avoid propagating any invalid orbital data. As seen in code 4.7, this is repeated over a set period of time, with a configured time step, to generate a time series of satellite positions and velocities for later use in calculations.

Listing 4.7: SGP4 propagation

```
        while currentTime < endTime:
        jd, fr = GetJulianDate(currentTime)

        for name, sat in satellites:
            e, r ,v = sat.sgp4(jd, fr)
            if e == 0:
                propagated[name].append({
                    "time": currentTime,
                    "r": r,  # position (km)
                    "v": v  # velocity (km/s)
                })
        currentTime += time_delta
```

The code seen in 4.7 generates a dictionary of satellite state vectors indexed by satellite name, with each entry containing a list of time stamped position and velocity vectors. An example of this is shown in Listing 4.8. This format allows for efficient and easy access to satellite positions, and their trajectories.

Listing 4.8: Structure of propagated satellite state vectors stored in a dictionary

```
propagated = {
    "SAT-44057": [
        {
```

```
            "time": datetime(2026, 4, 12, 12, 0, 0),
            "r": [-5321.84,  2154.77,  3219.65],  # km
            "v": [ 1.92, -6.48,  3.41]            # km/s
        },
        {
            "time": datetime(2026, 4, 12, 12, 1, 0),
            "r": [-5310.12,  1765.43,  3458.90],
            "v": [ 2.04, -6.51,  3.22]
        }
    ],

    "STARLINK-5694": [
        {
            "time": datetime(2026, 4, 12, 12, 0, 0),
            "r": [1034.55, -6882.21,  1421.88],
            "v": [ 7.32,   1.14,     0.98]
        }
    ]
}
```

4.4 Coordinate Transformations

Satellite positions produced by SGP4 are given in True Equator, Mean Equinox (TEME) format. TEME remains fixed to relative to the stars not to the Earth. It uses the Earths true equatorial plane at that moment in tim, but for the x-axis it uses the mean equinox which helps smooth out some of the short term wobbles in the Earths's rotation.

As TEME does not rotate with the Earth, but ground locations given in latitude and longitude do rotate with the Earth, for visibility and elevation calculations, both the satellite and the observer location need to be in the same rotating frame. For this reason TEME must be converted to Earth Centered Earth Fixed (ECEF) coordinate system [28].

4.4.1 TEME to ECEF

The conversion for TEME to ECEF is as follows ...

$$\mathbf{r}_{\text{ECEF}} = R_3(\theta_{\text{GMST}})\, \mathbf{r}_{\text{TEME}}, \tag{4.1}$$

Where:

- $R_3(\theta)$ is a rotation matrix describing a rotation about the Earth's Z–axis.
- θ_{GMST} is the Greenwich Mean Sidereal Time at the observation epoch.

This can be written explicitly as a rotation matrix as follows ...

$$R_3(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{4.2}$$

Meaning ...

$$\begin{bmatrix} x_{\text{ECEF}} \\ y_{\text{ECEF}} \\ z_{\text{ECEF}} \end{bmatrix} = \begin{bmatrix} \cos\theta_{\text{GMST}} & -\sin\theta_{\text{GMST}} & 0 \\ \sin\theta_{\text{GMST}} & \cos\theta_{\text{GMST}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{\text{TEME}} \\ y_{\text{TEME}} \\ z_{\text{TEME}} \end{bmatrix} \tag{4.3}$$

This aligns the satellite position with a Earth fixed rotating frame.

4.4.2 Geodetic ECEF

To compare a satellite with an observer on Earth, the observer's latitude, longitude and altitude must be expressed as cartesian X, Y, Z coordinates in the ECEF frame. As the Earth is not a perfect sphere it is approximated as a WSG-84 ellipsoid. This is defined by it's semi major axis(the equatorial size), the flattening(how much the poles are compressed) and the first eccentricity (a measure of how non spherical the ellipsoid is). This can be seen in (4.4) [29].

$$f = \frac{a-b}{a}, \tag{4.4}$$

where a is the semi-major axis and b is the semi-minor axis of the Earth. The first eccentricity is then given by

$$e^2 = 2f - f^2. \tag{4.5}$$

Given:

- Latitude: ϕ (radians)
- Longitude: λ (radians)
- Altitude: h (meters above the WGS–84 reference ellipsoid)

WGS–84 Ellipsoid Constants:

- $a = 6378137.0$ m (equatorial radius)
- $f = \dfrac{1}{298.257223563}$ (flattening)
- $e^2 = 2f - f^2$ (first eccentricity squared)

The first step is to find the distance from Earth's center to the surface at a given latitude accounting for any potential bulges at the equator. This can be seen in 4.6 [29]

$$N(\phi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \phi}}. \tag{4.6}$$

Once $N(\phi)$ has been found the observer location can be described in ECEF as seen in (4.7) [29].

$$X = (N(\phi) + h) \cos\phi \cos\lambda \tag{4.7}$$

$$Y = (N(\phi) + h) \cos\phi \sin\lambda \tag{4.8}$$

$$Z = \big(N(\phi)(1 - e^2) + h\big) \sin\phi \tag{4.9}$$

These equations convert the geodetic position (ϕ, λ, h) and the satellite's equation into the ECEF frame. This allows for all later geometry such as line of sight checks to be carried out.

4.5 Visibility Logic

Once both the ground observer and the satellites propagated position are represented in ECEF the first filter check for visibility can be applied. This is based on computing the satellite's elevation angle relative to the observers horizon.

Given the satellite position,

$$r^{\text{ECEF}}_{\text{satellite}} = (s_x, s_y, s_z), \tag{4.10}$$

and the observer position,

$$r^{\text{ECEF}}_{\text{observer}} = (g_x, g_y, g_z), \tag{4.11}$$

the line of sight vector (LOS) can be represented as,

$$d = r^{\text{ECEF}}_{\text{satellite}} - r^{\text{ECEF}}_{\text{observer}} = (d_x, d_y, d_z), \tag{4.12}$$

Using the observer latitude and longitude, the LOS can be converted into local horizon coordinates. This can be done following (4.13) [30]

$$\begin{aligned} e &= -\sin\lambda\, d_x + \cos\lambda\, d_y, \\ n &= -\cos\lambda \sin\phi\, d_x - \sin\lambda \sin\phi\, d_y + \cos\phi\, d_z, \\ u &= \cos\lambda \cos\phi\, d_x + \sin\lambda \cos\phi\, d_y + \sin\phi\, d_z. \end{aligned} \tag{4.13}$$

Using this, the elevation angle can then be computed as the angle between the LOS vector and the local horizon [30].

$$\theta_{\text{el}} = \arctan\left(\frac{u}{\sqrt{e^2 + n^2}}\right) \tag{4.14}$$

Then, to filter out any satellites which are too close to the horizon or below the horizon, a mask angle of 15° is applied.

$$\text{el} \geq \theta_{\text{mask}}, \qquad \theta_{\text{mask}} = 15^\circ \tag{4.15}$$

4.6 Beam width Coverage Model

At each time step, satellite and observer positions are represented in ECEF coordinate frame. If we let the satellite position be r_{sat} and the observer position be r_{obs}, and the nadir direction of the satellite is defined as a vector pointing from the satellite towards the Earth's centre be given by $-r_{sat}$. The direction from the satellite to the observer can be found with $r_{obs} - r_{sat}$. The angle between the satellite nadir direction and the observer direction can then be computed as seen in 4.16

$$\theta = \cos^{-1}\left(\frac{(-\mathbf{r}_{sat}) \cdot (\mathbf{r}_{obs} - \mathbf{r}_{sat})}{\|\mathbf{r}_{sat}\| \, \|\mathbf{r}_{obs} - \mathbf{r}_{sat}\|}\right) \tag{4.16}$$

A satellite is capable of providing server if this angle is less than or equal to the configured half power beam with of a satellite θ_{beam}.

The beam width parameter is configured within the system allowing different antenna characteristics to be evaluated without modifying the core implementation. This beam width check is executed for every visible satellite at each simulation time step. In this project the half power is set to be 7.5°, this is a liberal angle in order to attempt to account for beam forming techniques. [31]

$$\theta \leq \theta_{beam} \tag{4.17}$$

4.7 Coverage Window Detection and Tracking

To determine coverage for a given area, the system iterates over a grid of latitude and longitude points within the area of interest. For each point, at each time step, the visibility and beam width checks are applied to all propagated satellites. If one or more satellites satisfy both conditions, that grid point is marked as having coverage for that time step. The system tracks when coverage becomes available and when it is lost, allowing it to identify coverage windows. Each window is recorded with its start time, end time, duration, and the satellite(s) providing coverage during that window. This information is crucial for analyzing the availability of satellite service in the region. This logic can be seen implemented in code 4.9.

Listing 4.9: Visibility window logic

```
coverage_windows = []
in_coverage = False
window_start = None

for t, _, _, visible_satellites in visibility_by_time:

    filtered_satellites, _ = BeamFilter(
        visible_satellites, jd, fr, lat, lon,
        obs_alt=0.0, beamwidth_deg=beamwidth
    )

    coverage_available = len(filtered_satellites) > 0

    if coverage_available and not in_coverage:
        in_coverage = True
        window_start = t

    if not coverage_available and in_coverage:
        in_coverage = False
        coverage_windows.append((window_start, t))

# Close final window if coverage persists
```

```
if in_coverage:
    coverage_windows.append((window_start, visibility_by_time[-1][0]))
```

4.8 Coverage Capability

Coverage capability is measured in the theoretical maximum throughput that a satellite can provide to a given location using Shannon's capacity formula. This is calculated for each satellite that satisfies the visibility and beam width checks at each time step. The system assumes a fixed bandwidth and signal-to-noise ratio (SNR). This allows for an analysis of how much data throughput can be expected during those coverage windows.

$$P_{\text{received}}\ (\text{dBm}) = P_{\text{transmitted}}\ (\text{dBm}) + G_t\ (\text{dB}) + G_r\ (\text{dB}) - L_{\text{path}}\ (\text{dB}) - L_{\text{other}}\ (\text{dB}). \tag{4.18}$$

In Equation (4.18), P_{received} (dBm) is the power level at the receiver input after all gains and losses along the communication link have been applied. It represents the signal strength available for demodulation at the receiver.

$P_{\text{transmitted}}$ (dBm) is the output power of the transmitting radio or antenna system.

G_t (dB) is the transmitter antenna gain, which quantifies the ability of the transmitting antenna to direct radiated power toward the receiver relative to an isotropic radiator.

G_r (dB) is the receiver antenna gain, representing the effectiveness of the receiving antenna in capturing incoming electromagnetic energy from the direction of the transmitter.

L_{path} (dB) is theaptoror free-space path loss, accounting for signal attenuation due to geometric spreading of the electromagnetic wave as it propagates over the distance between transmitter and receiver.

L_{other} (dB) represents additional system losses not captured by free-space path loss alone, including atmospheric attenuation, polarization mismatch, pointing losses, hardware losses, and implementation-related inefficiencies.

Equation 4.18 (Link budget equation), calculates the received power, in dBm based on transmitted power, antenna gains, and losses in the communication path. This allows the power of a signal at a user's location to be calculated.

$$P_{\text{interference}} = 10\log_{10}\left(\sum_{i=1}^{N} 10^{\frac{P_i}{10}}\right). \tag{4.19}$$

Equation 4.19 [32] converts dBM into watts, sums all interfering signals received at a given location, and converts these interfering signals back into dB. This allows us to see the total interfering power.

$$P_{\text{noise}} = -174 + 10\log_{10}(B) + NF. \tag{4.20}$$

Equation 4.20 [33]calculates the thermal noise power at room temperature, which simply gives a baseline which can be used to estimate background noise which will limit how well a receiver could detect a signal. B is the channel bandwidth in Hz and NF is the noise figure which comes from the receiver hardware.

$$\text{SNR} = P_{\text{signal}} - P_{\text{noise}}. \tag{4.21}$$

Equation 4.21 [33]calculates the signal to noise ratio, this is a measure of the main signal's strength relative to background noise.

$$\text{SIR} = P_{\text{signal}} - P_{\text{interference}}. \tag{4.22}$$

Equation 4.22 [33] is similar to 4.20, except that it calculates the strength of the main signals relative to interference signals from nearby towers.

$$\text{SINR (dB)} = 10\log_{10}\left(\frac{10^{\frac{P_{\text{signal}}}{10}}}{10^{\frac{P_{\text{interference}}}{10}} + 10^{\frac{P_{\text{noise}}}{10}}}\right). \tag{4.23}$$

SINR is calculated as the main signal power divided by the sum of interference and noise. This is done in a linear scale (mV) and then converted back into dB.

Given these calculations, the Shannon capacity (theoretical maximum data rate) can be found. This can be seen in 4.24 [32]

$$Capacity = Bandwidth \cdot \log_2\left(1 + \text{SINR}_{\text{linear}}\right). \tag{4.24}$$

These equations were implemented in the system to calculate the theoretical maximum throughput for each satellite providing coverage at each time step. This allows for an analysis of coverage in Ireland.

Parameters for the constellations were taken from publicly available sources, and are summarized in Table **??**. These parameters were used to calculate the received power, interference, noise, and ultimately the Shannon capacity for each satellite providing coverage at each time step.

To calculate the EIRP for Starlink, the following calculations were preformed.

$$\begin{aligned} \text{Given:} \quad & \text{EIRP spectral density} = 17.9\text{–}21.5 \text{ dBW/MHz} \\ & B = 250 \text{ MHz} \end{aligned}$$

$$\text{Total EIRP (dBW)} = \text{EIRP density (dBW/MHz)} + 10\log_{10}(B)$$

$$10\log_{10}(250) = 23.98 \approx 24 \text{ dB}$$

$$\text{Lower bound:} \quad 17.9 + 24 = 41.9 \text{ dBW}$$

$$\text{Upper bound:} \quad 21.5 + 24 = 45.5 \text{ dBW}$$

$$\boxed{\text{Total EIRP} \approx 41.9\text{–}45.5 \text{ dBW}}$$

A similar calculation was preformed for the OneWeb constellation as the EIRP was not readily given.

Table 4.1: Link budget parameters used for LEO system modelling.

System	**Altitude (km)**	**Transmit Power (dBW)**	**Receiver Gain (dBi)**	**Frequency (GHz)**
Starlink (FSS) [34]	550	44–46	33	12
Starlink DTC [35]	300–550	∼58	∼0	2
Kuiper [36]	590–630	44–46	33	19
OneWeb [37]	1200	41–44	33	12

Notes: Starlink and Kuiper transmit powers represent equivalent downlink EIRP derived from reported spectral density values. Starlink Direct-to-Cell operation is constrained by regulatory power-flux-density limits rather than beam EIRP. OneWeb operates at higher altitude, resulting in increased path loss and reduced SNR.

4.9 Optimal Satellite Selection

At each simulation time step, multiple satellites may satisfy the visibility and beam width checks for a given location. To determine which satellite would provide the best service, the satellite whose beam is most closely aligned with he observer is selected as the serving satellite. Alignment is measured by the angular separation between the satellite's nadir direction and the line of sight to the observer. The satellite with the smallest angle is selected as the serving satellite for that time step.

Let the satellite position vector at a given time step be

$$\mathbf{r}_{\text{sat}} \in \mathbb{R}^3,$$

and the observer position, transformed into the same reference frame, be

$$\mathbf{r}_{\text{obs}} \in \mathbb{R}^3.$$

The nadir beam direction of the satellite is defined as

$$\mathbf{u}_{\text{beam}} = \frac{-\mathbf{r}_{\text{sat}}}{\|\mathbf{r}_{\text{sat}}\|}.$$

The line-of-sight vector from the satellite to the observer is

$$\mathbf{v}_{\text{LOS}} = \mathbf{r}_{\text{obs}} - \mathbf{r}_{\text{sat}},$$

with the corresponding unit vector

$$\mathbf{u}_{\text{LOS}} = \frac{\mathbf{v}_{\text{LOS}}}{\|\mathbf{v}_{\text{LOS}}\|}.$$

The angular separation between the beam centre and the observer direction is computed as

$$\theta = \cos^{-1}(\mathbf{u}_{\text{beam}} \cdot \mathbf{u}_{\text{LOS}}) \,.$$

A satellite satisfies the beam-width constraint if

$$\theta \leq \frac{\theta_{\text{beam}}}{2}.$$

For a given time step, the optimal satellite is selected as

$$s^* = \arg\min_{s_i} \; \theta_i,$$

where this is preformed over all satellites that are visible and capable of providing coverage at that time step.

4.10 Computational Complexity and run time

This sections aims to provide and overview of the dominant computational cost, scaling considerations and practical performance limitations. The overall run time of the system is influenced by several factors, including the number of satellites, the duration of the simulation, the time step size, and the resolution of the coverage grid.

We can let N_s be the number of satellites, N_t be the number of time steps, and N_g be the number of grid points. Orbit propagation using SGP4 has a computational complexity of $\mathcal{O}(1)$ per satellite per time step. This means, the total cost of orbit propagation is $\mathcal{O}(N_s \cdot N_t)$, as each satellite's position must be computed at each time step. For a large LEO constellation such as Starlink, with thousands of satellites and a simulation which spans multiple hours with a fine time step, this can lead to a significant computational load.

Coverage analysis adds another layer to this, at each time step, visibility and beam-width checks are preformed for every satellite-grid point combination. This can result in a complexity of $\mathcal{O}(N_s \cdot N_t \cdot N_g)$. Even though each indivudual operation, such as the coordinate transformations, angle calculations and coverage check are relatively simple and inexpensive, the sheer number of operations can lead to long run times, especially as the grid resolution increases. For example, if we have 1000 satellites, a simulation duration of 24 hours with a time step of 10 seconds (which means 8640 time steps), and a grid of 1000 points, the number of operations would be $1000 \cdot 8640 \cdot 1000 = 8.64 \times 10^9$ operations, which take hours to compute.

To reduce the excessive computational costs, some design choices were implemented. FIrst, TLE data is loaded and parsed once at the beginning of the simulation and stored in memory, this avoids repeat network access or file reads which can take time. Secondly, the satellite propagation results are stored as time-indexed state vectors, allowed easy and repeated use during coverage checks without needing to re-propagate. The satellites which produce error codes at a given time step are also discarded to avoid unnecessary calculations. Finally, at any given time step at a grid point, if no satellites are visible, coverage checked are skipped entirely, and automatically marked as no coverage, this avoids the need to preform beam width checks for any satellites which are not even visible.

However, this creates issues with memory usage, as the system needs to store the propagated state vectors for all satellites at all time steps. For a large constellation and long simulation duration, this can lead to high memory consumption. To get around this, larger satellite constellations are ran for 12 hour periods instead of 24, with time steps of 30 seconds instead of 10 or 15. This increases the speed of the simulation by 2-3x.

5. Results

The three core metrics which will be compared are coverage availability (as a percentage of time), capacity throughput (Mbps) and latency (ms).

Coverage availability determines how often the user has service, this metric is based on visibility of satellites and a high level model of the beamwidth of a given satellite. This will be compared to urban and rural 5G networks. Throughput is measured in Mbps, this is averaged over a period of time and will also be compared to urban and rural 5G throughput.

Number of satellites visible over time

Plots (elevation vs. time, count of visible satellites, etc.)

Sample day simulation

Visualisations

Coverage heatmaps

Capacity throughput estimates

5.1 5G Coverage Analysis

5.2 Performance Metrics

5.2.1 Throughput

(a) 5G coverage in Ireland – 48 [38]

(b) 5G coverage in Ireland – Eir [38]

(c) 5G coverage in Ireland – Three [38]

(d) 5G coverage in Ireland – Vodafone [38]

Figure 5.1: Network operator 5G coverage maps

6. Conclusions

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